

Supporting Information

Toughened bi-modal cathodes for all-solid-state batteries via controlled interfacial heterogeneity

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Supplementary Note 1. Electrochemical-mechanical modeling

A 2D electrochemical model integrated with a mechanics module was developed to predict the stress field in NCA particles during charging. It was assumed that the NCA particle is fully discharged in its initial state, and has a uniform Li concentration throughout its volume. To reduce the computational burden, diffusion of Li ions and intercalation-induced strain are assumed to be isotropic. During charging, the local current density (i_{loc}) at the NCA/LPSCl interface is determined by the Butler-Volmer equation (eq S1):

$$i_{\text{loc}} = -i_0 \left(\exp\left(\frac{\alpha_a F \eta}{RT}\right) - \exp\left(\frac{-(1-\alpha_a) F \eta}{RT}\right) \right) \quad (\text{S1})$$

where i_0 is the exchange current density, α_a is the transport coefficient, F is the Faraday constant, and η denotes the overpotential. R and T refer to the molar gas constant and the absolute temperature, respectively.

Solid-phase diffusion of intercalated Li in an NCA particle follows Fick's law (eq S2):

$$\frac{\partial c}{\partial t} - D \nabla^2 c = 0 \quad (\text{S2})$$

where c is the solid-phase concentration of Li in NCA, D is the diffusion coefficient, and t represents the time.

The stress-strain relationship of a linearly elastic material is described by eq S3:

$$\varepsilon_{ij}^e = \frac{1}{E} [(1 + \nu) \sigma_{ij} - \nu \sigma_{kk} \delta_{ij}] \quad (\text{S3})$$

where ε_{ij}^e and σ_{ij} represent strain and stress tensors, respectively. E is the Young's modulus, ν is the Poisson's ratio, and δ_{ij} is the Kronecher delta. Accordingly, the total strain (ε_{ij}), including the diffusion-induced strain (ε_{ij}^d) arising from the lithiation/delithiation of NCA can be expressed by eq S4:

$$\varepsilon_{ij} = \varepsilon_{ij}^e + \varepsilon_{ij}^d = \frac{1}{E} [(1 + \nu) \sigma_{ij} - \nu \sigma_{kk} \delta_{ij}] + \frac{\Omega}{3} \Delta c \quad (\text{S4})$$

where Ω is the partial molar volume of NCA and c is the Li concentration.

To simulate interfacial debonding behavior, a cohesive zone model is employed. Cohesive elements with zero thickness are inserted along the interface, ensuring initial mechanical continuity between adjacent objects. Debonding is assumed to follow a bilinear traction-separation law and initiates when the interfacial stress induced by volume change of NCA exceeds the cohesive strength. Damage initiation is governed by the quadratic nominal stress criterion, which is expressed by eq S5:

$$\left(\frac{\langle\sigma_t\rangle}{\sigma_t^c}\right) + \left(\frac{\sigma_s}{\sigma_s^c}\right) = 1 \quad (\text{S5})$$

where σ_t and σ_s represent the tensile and shear tractions in cohesive elements, and σ_t^c , σ_s^c are their respective critical values. The Macaulay bracket, denoted by $\langle \rangle$, is defined as $\langle x \rangle = x$ for $x > 0$ and $\langle x \rangle = 0$ for $x \leq 0$, thereby excluding compressive normal stresses from contributing to damage initiation. Under mixed-mode loading, the effective separation displacement d_m is given by eq S6:

$$d_m = \sqrt{\langle d_n \rangle^2 + d_s^2} \quad (\text{S6})$$

where d_n and d_s denote the relative displacements in normal and shear directions, respectively. Prior to damage initiation, the traction–separation relationship is expressed by eq S7:

$$\sigma_i = K_i d_i \quad (\text{S7})$$

where K denotes the stiffness of the cohesive elements, and the subscript i indicates the directional component, either normal or shear. The critical mixed-mode displacement d_{m0} at which damage initiates is determined by eq S8:

$$d_{m0} = \begin{cases} d_n^0 d_s^0 \sqrt{\frac{d_m^2}{(d_s^0 d_n)^2 + (d_n^0 d_s)^2}} & d_n > 0 \\ d_s^0 & d_n < 0 \end{cases} \quad (\text{S8})$$

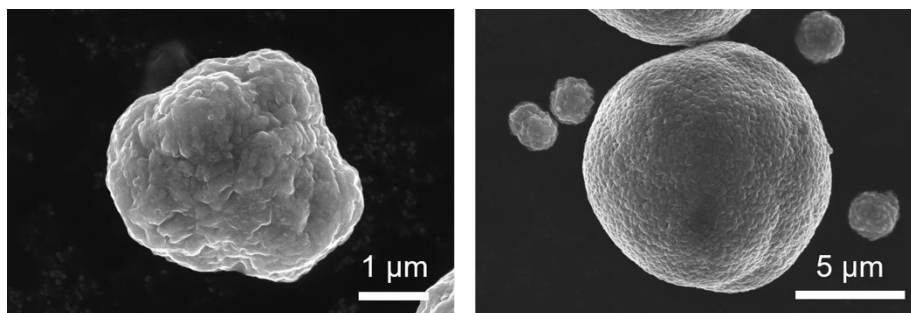


Figure S1. SEM images of small and large NCA particles in B-NCA cathodes, diameter of which are 3 and 10 μm , respectively.

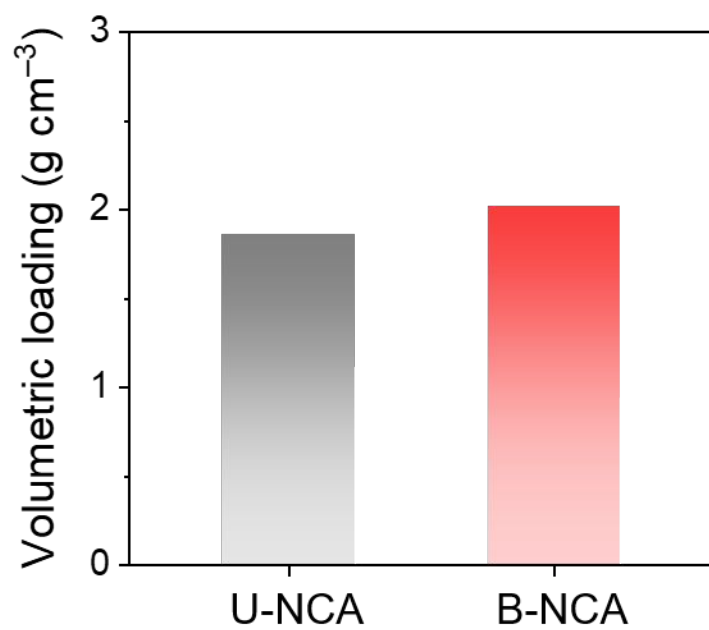


Figure S2. Volumetric loading of composite cathodes with U-NCA and B-NCA.

Table S1. Fitted parameters of the EIS spectra obtained from cathode symmetric cells.

Cathode	$R_{\text{ion, bulk}}$ ($\Omega \text{ cm}^2$)	$R_{\text{ion, gb}}$ ($\Omega \text{ cm}^2$)	R_{ele} ($\Omega \text{ cm}^2$)
U-NCA	10.5	56.9	89.9
B-NCA	7.4	52.4	20.0

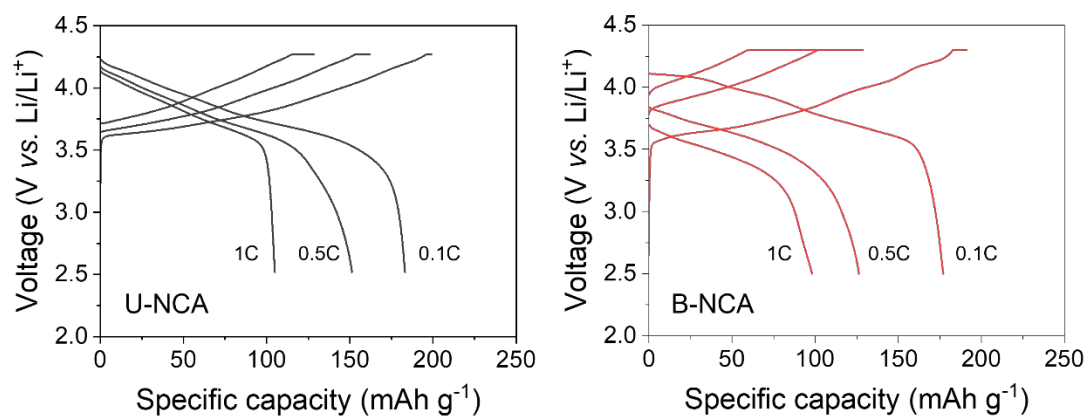


Figure S3. Rate performance tests of NCA | LPSCl | Li-In cells with U-NCA and B-NCA.

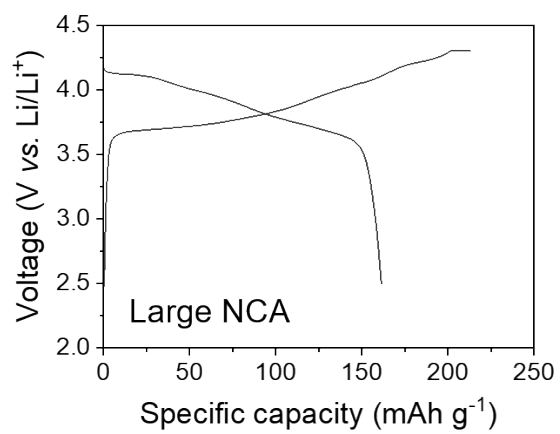


Figure S4. Voltage profile of the first cycle (0.1C) for an NCA || Li-In cell employing large (10 μm) NCA particles.

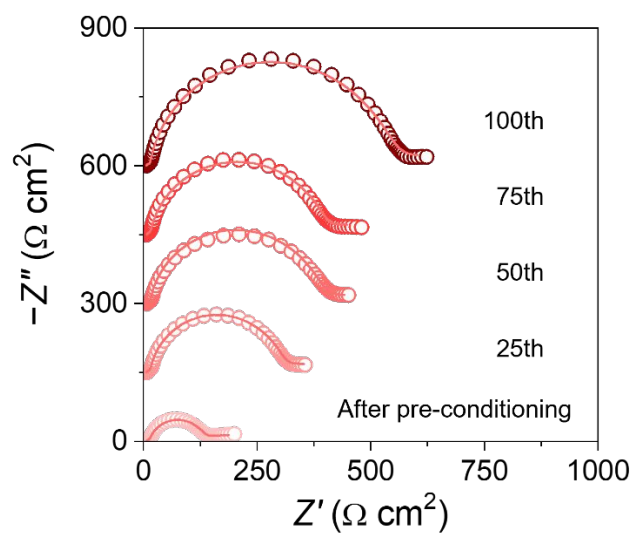


Figure S5. AC-impedance spectra of B-NCA | LPSCI | Li-In cells measured every 25 cycles at SOC = 100%.

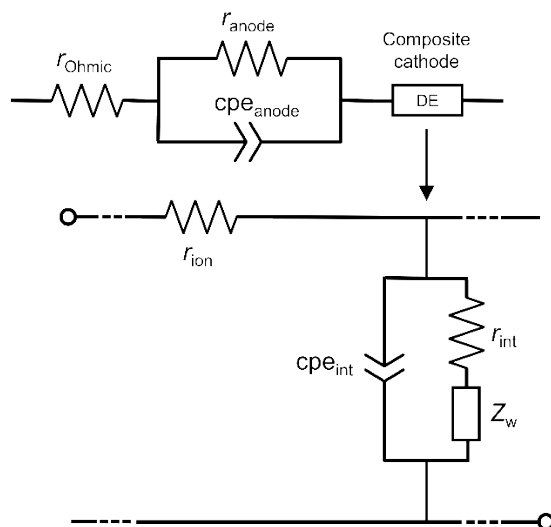


Figure S6. The equivalent circuit model used to fit the Nyquist plots of NCA | LPSCl | Li-In full cells. r_{ion} and r_{anode} in the equivalent circuit represent the ionic transport resistance within the composite cathode and the interfacial resistance at the Li-In/LPSCl interface, respectively. The interface between NCA and LPSCl is described by r_{int} , cpe_{int} , and z_w , denoting the interfacial resistance, constant-phase element, and Warburg resistance, respectively.

Table S2. Fitted parameters of the EIS spectra obtained from B-NCA | LPSCI | Li-In full cells during cycling at 0.5C.

Cycle number	R_{Ohmic} ($\Omega \text{ cm}^2$)	R_{anode} ($\Omega \text{ cm}^2$)	R_{ion} ($\Omega \text{ cm}^2$)	R_{int} ($\Omega \text{ cm}^2$)	R_{w} ($\Omega \text{ cm}^2$)	CPE_{int}	
						C ($\mu\text{F s}^{\eta-1} \text{cm}^{-2}$)	η (-)
1	16.7	13.4	5.7	113.5	70.7	112.6	0.84
25	17.7	15.1	6.0	275.6	106.0	29.7	0.91
50	17.7	15.6	6.6	332.2	176.7	28.3	0.91
75	17.7	1.14	6.7	335.7	212.0	21.2	0.90
100	17.7	15.6	6.7	501.8	353.4	14.2	0.90

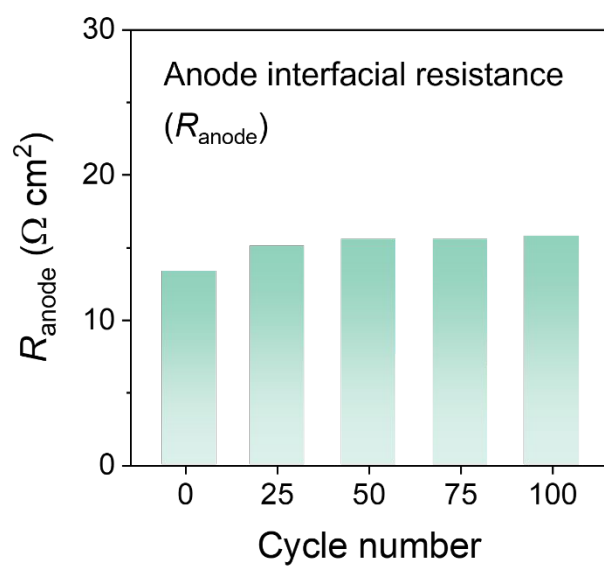


Figure S7. Quantified R_{anode} values the of B-NCA | LPSCl | Li-In cell measured every 25 cycles.

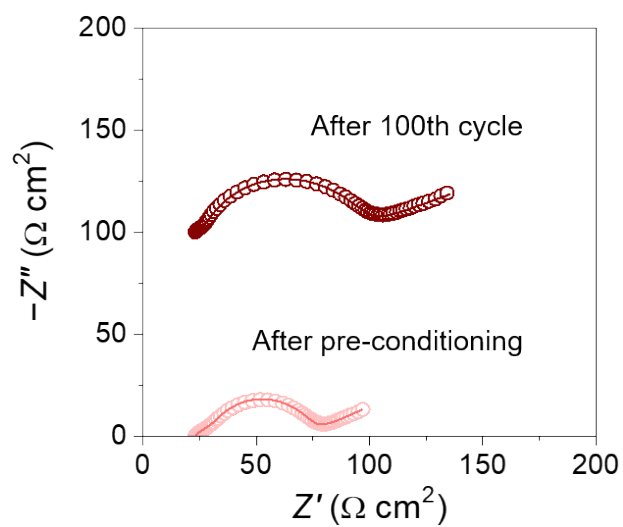


Figure S8. AC-impedance spectra of U-NCA | LPSCl | Li-In cells measured after pre-conditioning and 100th cycle at SOC = 100%.

Table S3. Fitted parameters of the EIS spectra obtained from U-NCA | LPSCI | Li-In full cells during cycling at 0.5C.

Cycle number	R_{Ohmic} ($\Omega \text{ cm}^2$)	R_{anode} ($\Omega \text{ cm}^2$)	R_{ion} ($\Omega \text{ cm}^2$)	R_{int} ($\Omega \text{ cm}^2$)	R_{w} ($\Omega \text{ cm}^2$)	CPE_{int}	
						C ($\mu\text{F s}^{\eta-1} \text{ cm}^{-2}$)	η (-)
1	20.7	8.0	10.4	56.0	59.5	29.5	0.86
100	21.3	8.6	11.2	84.5	123.4	24.7	0.81

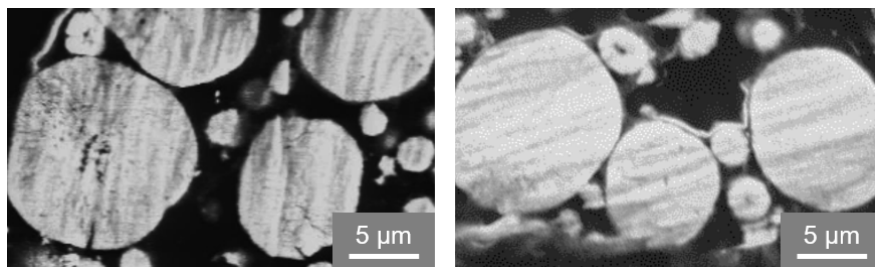


Figure S9. Cross-sectional SEM images of the composite cathodes after cell fabrication.

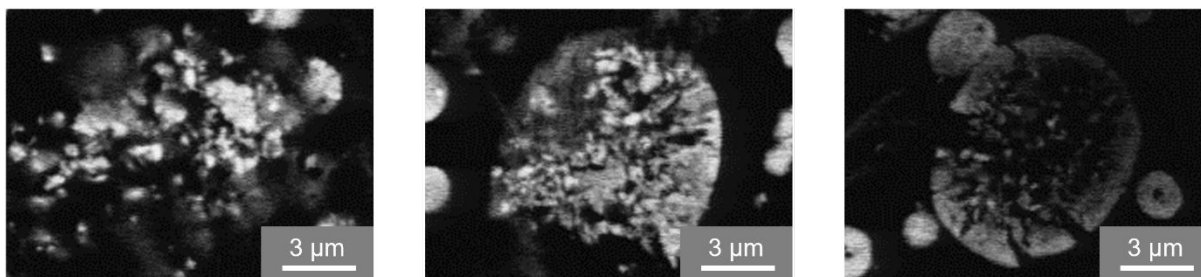


Figure S10. Cross-sectional SEM images of the cracked large NCA particles in the B-NCA cathode collected after 100 cycles.

Table S4. Parameters for electrochemical modeling.

Parameter [unit]	value
Cell dimension [L×W, μm]	815×100
Temperature [K]	298.15
Reference exchange current density [A m^{-2}]	10
Reference concentration [mol m^{-3}]	48000 [S1]
Radius of the active material [μm]	5
Maximum cell voltage [V]	4.3
Minimum cell voltage [V]	2.5
C-rate	0.33
Electrical conductivity of active material [S m^{-1}]	1
Diffusion coefficient of Li in active material [$\text{m}^2 \text{s}^{-1}$]	3×10^{-14} [S1]
Electrolyte conductivity [S m^{-1}]	0.02 [S2]
Interfacial strength [MPa]	100 [S3]
Critical energy release rate [J m^{-2}]	1 [S4]

Table S5. Boundary conditions for electrochemical modeling.

	Mass conservation	Charge conservation	Solid mechanics
Boundary 1	$\mathbf{n} \cdot \nabla c = 0$	$\mathbf{n} \cdot \nabla \phi_e = 0,$ $\mathbf{n} \cdot \nabla \phi_s = i_{\text{total}}$ $\phi_s = 0,$	$\mathbf{n} \cdot \sigma_{yy} = -P_{\text{app}}$
Boundary 2		$\mathbf{n} \cdot \nabla \phi_e = 0$	$\mathbf{n} \cdot \mathbf{u} = 0$
Boundary 3			
Boundary 4		$\mathbf{n} \cdot \nabla \phi_e = 0,$ $\mathbf{n} \cdot \nabla \phi_s = i_{\text{total}}$	
Boundary 5	-	$\mathbf{n} \cdot \nabla \phi_s = 0$	-
Electrolyte-active material interface	$-\mathbf{n} \cdot D \nabla c = \frac{i_{\text{loc}}}{aF}$	-	-

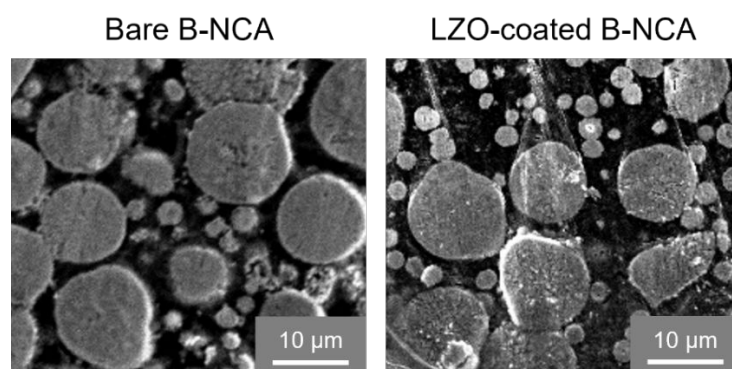


Figure S11. Cross-sectional SEM images of the composite cathodes employing bare B-NCA and LZO-coated B-NCA active materials.

Table S6. Fitted parameters of the EIS spectra obtained from LZO-coated B-NCA | LPSCI | Li-
In full cells during cycling at 0.5C.

Cycle number	R_{Ohmic} ($\Omega \text{ cm}^2$)	R_{anode} ($\Omega \text{ cm}^2$)	R_{ion} ($\Omega \text{ cm}^2$)	R_{int} ($\Omega \text{ cm}^2$)	R_{w} ($\Omega \text{ cm}^2$)	CPE_{int}	
						C ($\mu\text{F s}^{\eta-1} \text{ cm}^{-2}$)	η (-)
1	15.7	7.1	5.9	36.8	58.7	141.5	0.71
50	21.9	8.6	9.3	73.0	76.5	127.3	0.68
100	20.4	8.9	10.2	102.5	77.7	113.2	0.72

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